

Set 5: Kinetic theory

Notes

Set 5: Exercises

1. In Experiment 7 you examined the relationship between the pressure and volume of a gas as described by Robert Boyle. State Boyle's Law, give a mathematical relationship between pressure and volume and sketch a graph to show this relationship.
2. Aerosol cans, including deodorant sprays, have labels warning against exposing used cans to excessive heat. Record the warning label of an aerosol can that you have at home. Explain the reasons behind these warnings.
3. A car or bicycle tyre becomes hotter during use. Assuming the volume of the tyre remains constant what happens to the pressure of air inside the tyre? Explain in terms of the kinetic theory of matter.
4. A helium balloon when released rises up into the air and disappears out of sight.
 - (a) Why does the helium balloon rise?
 - (b) Describe the pressure changes both in and around a helium balloon as it rises. Use diagrams to help your explanation.
 - (c) What will eventually happen to the balloon? Explain.
5. Explain the boiling point of a liquid. Why does water boil at a temperature lower than 100°C at 2 000 m above sea level?
6. Explain vapour pressure of a liquid. Why is the boiling point of a solution higher than the boiling point of its solvent?